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1. Study Summary

Study Summary Table	
Project Title	**Modern Middle Eastern Genomics Project**
Objectives (brief overview)	The objective of this project is to reconstruct population histories in the Middle East through Whole Genome Sequenced human DNA. This modern DNA will be sequenced from saliva samples. The research questions will examine Lebanese, Syrian, and Mesopotamian population dynamics, admixture events, and pattern migrations, both in the modern and premodern periods. The populations in which saliva samples will be acquired will be individuals of Middle Eastern descent (adults only) living in the United States. Specifically, descended from Lebanese, Syrian, and Iraqi populations. The source of the specimens will be saliva samples. The acquisition of the saliva samples will ideally occur over the Summer of 2026.
Study collaboration	☒ Single-site ☐ Multi-site ☐ Collaborative ☐ Other (describe):
Interventions and research interactions (only list protocol driven procedures)	Saliva Collection
Length of involvement for individual participants	30-60 minutes
Research design	Users will be contacted and answer survey questions to determine eligibility for the study. If eligible, a saliva sample will be acquired for the user. All of the saliva samples will then be taken to the lab, extracted, and sequenced.
Key abbreviations and definitions	

2. Objectives

Objective Narrative
The objective of this project is to reconstruct population histories in the Middle East through Whole Genome Sequenced human DNA. This modern DNA will be sequenced from saliva samples. The research questions will examine Lebanese, Syrian, and Mesopotamian population dynamics, admixture events, and pattern migrations, both in the modern and premodern periods. The

populations in which saliva samples will be acquired will be individuals of Middle Eastern descent (adults only) living in the United States. Specifically, descended from Lebanese, Syrian, and Iraqi populations. The source of the specimens will be saliva samples. The acquisition of the saliva samples will ideally occur over the Summer of 2026.

3. Background

Background Narrative
The research will be covering a relatively underrepresented region of modern DNA. The only major study of this region was performed in "Genome-Wide Diversity in the Levant Reveals Recent Structuring by Culture" (Haber et al. 2013). Other than this, no major studies have been done for internal/external Lebanese, Syrian, and Iraqi population dynamics. Furthermore, this collected data will provide new information on our knowledge of population dynamics, migrations, and patterns in the modern and pre-modern Levant. The research team has experience in modern population genetics and downstream analysis using bioinformatics.

4. External Collaborators

External Sites and Collaborators			
List all external sites and independent, non-Emory investigators that are engaged in the research			
Name of External Site/ Organization	Site's Principal Investigator (PI)	Site PI's Email	Reviewing IRB
			☐ Emory ☐ Local IRB
			☐ Emory ☐ Local IRB
			☐ Emory ☐ Local IRB
			☐ Emory ☐ Local IRB
			☐ Emory ☐ Local IRB

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External Collaborators and Sites

5. Study Endpoints

Endpoint Summary	
Study Endpoints	
Primary	Successful collection and generation of usable genomic DNA data from participant saliva samples for population genetic analysis.
Secondary	Representation of targeted regional/ethnic groups in the dataset
Safety Endpoints	
• Incidence of participant discomfort, distress, or withdrawal related to saliva collection or survey questions • Any breaches of confidentiality or data security • Any participant complaints or adverse reactions during sample collection	

Endpoint Narrative
The primary endpoint of this study is the successful collection of saliva samples and generation of high-quality genomic DNA data suitable for downstream population genetic and bioinformatic analyses. Success will be determined based on DNA extraction yield, purity metrics, and sequencing or genotyping quality control standards.

6. Procedures Involved

Clinical Impact	
Does enrollment in this study potentially inform, modify, or change a patient's care? *Examples: length of a clinical intervention is extended, additional imaging is completed, a new clinical workflow is applied, a specific clinical standard is chosen/used with the patient*	☐ Yes ☒ No

Procedures Narrative
1. Study Design and Interventions
This study is an observational, non-interventional population genetics study involving the collection of saliva samples and demographic/ancestry information from adult participants of Middle Eastern heritage. The study does not involve clinical treatment, diagnosis, or medical decision-making, and participation will not influence or alter any aspect of the participant's healthcare.

Participants will be recruited through community outreach, advertisements, and/or word of mouth. Interested individuals will complete an eligibility screening and, if eligible, will undergo an informed consent process prior to participation.

Study procedures include:

- Completion of a demographic and ancestry questionnaire (approximately 30 minutes)
- Collection of one saliva sample using a commercially available saliva collection
- Saliva collection is a non-invasive procedure and does not deviate from standard use of saliva self-collection kits. No drugs, medical devices, or clinical interventions are used in this study.

The study involves a single study interaction. There are no follow-up clinical visits or procedures. To minimize risk, participants will be provided clear written instructions for saliva collection, and they may stop participation at any time. The research team will monitor for any participant-reported discomfort or concerns during the process.

2. Data and Specimens Collected

The following data will be collected directly from participants:

- Demographic information (e.g., age range, sex, country of origin, ancestry background)
- Self-reported family geographic origins
- Optional information about languages spoken or cultural affiliation (if applicable)

Data will be collected through secure electronic surveys or paper forms that are later entered into a secure database. To reduce the likelihood of erroneous or automated ("bot") responses for online surveys, the study will use measures such as CAPTCHA, duplicate entry checks, and manual review of responses for inconsistencies.

No medical records will be accessed, and no protected health information (PHI) from clinical records will be collected as part of this study.

One saliva sample will be collected per participant using a self-collection saliva kit. Saliva samples will be used for DNA extraction and subsequent genetic analyses related to population history and ancestry. Testing will include standard genomic laboratory procedures such as DNA extraction, quantification, and sequencing or genotyping. No clinical diagnostic testing will be performed, and individual genetic results will not be returned to participants.

3. Long-Term Follow-Up Plans

There are no in-person or clinical follow-up visits planned after the initial study procedures.

Participants may be contacted in the future only if they have provided optional consent to be re-contacted for related research studies. Any future contact would require separate IRB review and, if applicable, additional consent.

The study does not include tracking of medical outcomes, survival, or linkage to public records (e.g., vital statistics). Long-term use of collected data and specimens will be limited to research on population genetics and human history, as described in the consent form.

7. Study Timeline

Study Timeline Narrative
1. Study-Wide Timeline
Participant recruitment and enrollment are expected to occur over a few months following IRB approval. Enrollment will remain open until the target sample size is reached or the recruitment window ends, whichever comes first. All study procedures for individual participants (consent, questionnaire, and saliva collection) occur during a single interaction, so there are no sequential sub-studies or dependent research phases requiring staggered completion. Laboratory processing of saliva samples (DNA extraction and quality control) will occur on a rolling basis as samples are received and is expected to be completed within 3–4 months after the final sample is collected. Primary genetic and population-level analyses are expected to be completed within 12 months after the close of enrollment**.** No study procedures are dependent on findings from earlier sub-studies, and no interim analyses will affect ongoing participant involvement.
2. Participant-Specific Time Commitment
Each participant will complete one study interaction. This will not take more than 30 minutes.

Schedule of Assessments

	Screening (phone) Day -7 to -1	**Saliva Collection (Single Visit)**
Participant Activities		
Screening consent and eligibility questions	X	
Demographics	X	
Informed Consent		X
Procedure/Saliva Collection		X

Footnotes/Key	
X	To be completed
(I)	Completed only if clinically indicated
Ph	May be completed over the phone
--XXX--	Continuous activity continuing between visit

8. Analysis Plan

Analysis Plan Narrative
The analysis will be performed over Whole Genome Sequenced human data. The data will be aligned to a reference genome, converted into analyzable genome formats (such as plink format or eigenstrat) and then analyzed using PCA, admixture analysis using ADMIXTOOLS, and f statistics. Additional computational genomic and statistical methods may be developed in the process of downstream analysis.

9. Population

Population Narrative
The research questions will examine Lebanese, Syrian, and Mesopotamian populations. Therefore, the saliva acquisition will target individuals from these regions. Ideally 80-90 samples/individuals will be acquired in diasporic communities in the United States. Features such as population distribution, religious affiliations, genetic continuity from ancient samples, and internal migrations will be studied among these populations. Other questions will examine admixture events in these populations in the pre-modern period, which can be determined in Whole Genome sequenced individuals. As a diasporic member in one of these communities, I will be able built strong community-based rapport.

10. Inclusion and Exclusion Criteria

Inclusion/Exclusion Criteria Narrative
1. General Eligibility Criteria
Inclusion Criteria: • Adults aged 18 years or older • Self-identify as having Middle Eastern ancestry (e.g., individual, parent, or grandparent born in a Middle Eastern country) • Able to understand the consent information and provide informed consent in English (or other approved study language, if applicable) • Willing and able to provide a saliva sample and complete a demographic/ancestry questionnaire Exclusion Criteria: • Individuals under 18 years of age

- Individuals who do not meet the ancestral criteria.

• Individuals unable to provide informed consent due to cognitive impairment or other decision-making limitations
• Individuals who are currently incarcerated or otherwise classified as prisoners (as defined by federal regulations)
• Individuals who have a medical or oral condition that prevents safe saliva sample collection (e.g., severe dry mouth disorders)

Describe how will eligibility be determined:

Eligibility will be determined in the phone screening.

11. Projected Enrollment

Enrollment Summary Narrative
The anticipated phone screening failure rate is low since contact with a strong community has been established. There are no study sub-groups (apart from different regions) and the demographic breakdown will not be determined by sex/gender but rather regional origins in the aforementioned targeted regions.

12. Advertising, Recruitment, and Screening

Recruitment Methods Narrative
1. General Recruitment Overview
All recruitment will be targeted and performed by contacting community members via the phone. Most of the recruitment will be performed on a word of mouth basis.
2. Specialized Recruitment Requirements
N/A
3. Targeted Recruitment Details
N/A

Participant Screening Narrative		
1. Screening Methods Overview		
The screening will be a set of questions on the phone asking regional origins, parental/grandparental regional origins.		
2. Screening Consent Requirements		
	a. Is a screening consent used or required for this study?	☐ Yes → *Complete the rest of the section* ☒ No → *Move to next section*
	b. Will HIPAA Authorization be added to the screening consent?	☐ Yes ☒ No
	c. Indicate how consent for screening will be obtained (Select all that apply)	☐ Signed, in-person/physical signature ☐ Signed, electronically (i.e., e-signature) ☒ Verbal (waiver requested) ☐ Online, without signature (waiver requested)
	d. Briefly describe the screening consent process	

13. Informed Consent

General Informed Consent Narrative
1. Informed Consent Overview
The consent process will be performed via a digital contract before acquiring saliva. The consent narrative will be the following: **1. Informed Consent Overview** Responsible Parties The informed consent process will be conducted by trained members of the research team, such as the Principal Investigator, study coordinator, or other designated research staff who have completed required human subjects protection training. **Timing** The consent process will occur **before any study procedures begin**. There will be no mandatory waiting period between introducing the study and obtaining consent; however, participants will be encouraged to take as much time as they need to review the information and ask questions. Participants will have access to the consent document **before and during** the consent discussion. The consent process is expected to take approximately **10–15 minutes**, depending on participant questions.

Location and Format

The consent process will take place **either in person or remotely**, depending on recruitment method:

- **In-person consent:** Participants will review a printed consent form with a study team member present to guide the discussion and answer questions.
- **Remote consent (if applicable):** Participants will review the consent document electronically with the opportunity to discuss the study with a research team member by phone or video call.

The consent process is **synchronous**, meaning a study team member will be available in real time to review the study and answer questions.

Participants will receive a **copy of the signed consent form** for their records, either as a printed copy (in person) or a downloadable/email copy (electronic consent).

Electronic Consent (if used)

If eConsent is used, it will be administered through a secure, institutionally approved electronic platform that allows participants to review the consent document and provide an electronic agreement. The system will document the date/time of consent and provide participants with a copy for their records.

Participants will not have to provide a signature but rather agrea via a clickwrap agreement.

2. Intentional Consent Practices

Intentional Consent Practices

Ensuring Participant Understanding

Study staff will explain the study purpose, procedures, risks, benefits, and voluntary nature of participation in plain language. Participants will be encouraged to ask questions at any time. To confirm understanding, study staff may ask participants to summarize key aspects of the study (e.g., what participation involves and that genetic results will not be returned) before proceeding.

Minimizing Coercion or Undue Influence

Participation is completely voluntary. Recruitment materials and consent discussions will clearly state that declining or withdrawing from the study will involve **no penalty or loss of benefits**. The study does not involve recruitment of individuals in dependent relationships with the research team (e.g., students or employees under the supervision of the PI). No excessive compensation will be offered.

Ongoing Consent

Because this study involves a single interaction, ongoing consent will be ensured by checking in with participants throughout the visit. Participants will be reminded that they may skip any questions, decline to provide a saliva sample, or stop participation at any time.

Deception or Incomplete Disclosure

This study does **not** involve deception or incomplete disclosure. Participants will be fully informed about the nature of the genetic research, how their data and samples will be used, and the limits of confidentiality.

14. Reporting Results and Incidental Findings

Participant Results Reporting Narrative
1. Routine Results
This study is designed for research purposes only and does not involve clinical or diagnostic testing. Laboratory analyses will focus on population-level genetic variation and ancestry patterns, not on identifying medically relevant genetic variants. Because testing will not be conducted in a CLIA-certified clinical laboratory and is not intended for medical interpretation, no individual research results will be returned to participants. This includes ancestry estimates, genetic variants, or other genomic findings. The consent process will clearly explain that the research is not a substitute for clinical genetic testing and that participants should consult a healthcare provider if they have concerns about their health or family history.

Participants will be informed of this results policy during the consent process and will acknowledge their understanding before enrollment.
2. Incidental Results
There are no incidental result risks as the data will be purely analyzed for population genetics and not for disease related data.

Study-Wide Result Reporting Narrative
Participants may be offered access to a general summary of study findings describing overall patterns of genetic diversity and population history identified in the study. These summaries will not include any individual-level data. Study findings may be presented at scientific conferences and published in peer-reviewed journals focused on genetics, anthropology, or human population history. Data shared publicly will be de-identified and comply with all applicable ethical and data-sharing regulations. Where appropriate, study findings may be shared with Middle Eastern community groups, cultural organizations, or educational forums in a general, non-technical format. These efforts will focus on broad insights into regional population history rather than individual genetic information.

15. Withdrawal of Participants

Withdrawal Narrative
1. Researcher-Initiated Withdrawals
Participants may be withdrawn from the study by the research team without their consent under limited circumstances, including: • The participant is found to be ineligible after enrollment (e.g., under 18 years of age, does not meet ancestry criteria, or unable to provide informed consent) • The participant is unable to safely provide a saliva sample • The participant engages in behavior that disrupts study procedures or poses a safety concern for staff If a participant is withdrawn by the research team, they will be informed of the reason for withdrawal by a study team member. Because this study involves no medical interventions, there are no medical follow-up procedures required after withdrawal. If withdrawal occurs before a saliva sample or questionnaire data are collected, no research data will be retained. If withdrawal occurs after data or a sample have been collected, participants will be

given the option to request destruction of their identifiable sample and associated data, to the extent possible. Data that have already been de-identified and incorporated into aggregate analyses may not be removable.
2. Participant-Initiated Withdrawals
Participation in this study is entirely voluntary. Participants may withdraw at any time, for any reason, without penalty. If a participant chooses to withdraw before completing study procedures, no further data or specimens will be collected. If withdrawal occurs after a saliva sample or questionnaire data have been provided, participants may request that their remaining identifiable samples and data be destroyed and not used for future research, where feasible. Because this study involves a single interaction and no ongoing interventions, there are no partial withdrawal procedures related to follow-up visits or continued monitoring. However, participants may choose to: • Skip any questionnaire questions • Decline to provide a saliva sample • Decline optional future contact Withdrawing from optional future contact does not affect participation in the current study.

16. Risks and Burdens

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Risks to Participants Narrative
1. Risks, Discomforts, and Burdens
This study involves no risk to participants. Saliva collection is non-invasive and has no discomforts.
2. Steps to Minimize Risks
n/a

17. Potential Benefits to Participants

Benefits Narrative

Participants will not receive any direct medical or personal benefit from participating in this study. The genetic analyses conducted are for research purposes only and are not designed to provide health information, clinical diagnoses, or individualized ancestry reports. Some participants may find personal value in contributing to research on Middle Eastern population history and genetic diversity. However, this is not considered a direct benefit of participation.

18. Data and Safety Monitoring

Data Monitoring Requirements Assessment	
1. Do any of the below, highest-complexity categories apply to the research? *(i.e., did you answer "yes" to 1a or 1b?)*	☐ Yes → *This is a complexity cat. A study, insert table 1* ☒ No → *Go to #2*
a. **This a Phase I/II/III Clinical Trial with an IND or significant risk IDE**	☐ Yes ☐ No
b. **The study or trial includes high-risk procedures**	☐ Yes ☐ No
2. Do any of the below, high-complexity categories apply to the research? *(i.e., did you answer "yes" to one of items listed in 2a-2d?)*	☐ Yes → *This is a complexity cat. B study, insert table 1* ☐ No → *Go to #3*
a. **Study or trial is expected to be IND Exempt, IDE Exempt, or under an Abbreviated IDE/Non-Significant Risk Device**	☐ Yes ☐ No
b. **Clinical trial of drugs or devices used under their FDA-approved indication** (e.g., comparative effectiveness trial of standard of care interventions)	☐ Yes ☐ No
c. **Application of software or algorithm that will inform clinical care or direct care interventions**	☐ Yes ☐ No
d. **Application of other novel clinical techniques or intervention** (e.g., nonstandard surgical step)	☐ Yes ☐ No
3. If none of the above categories apply, insert the medium-complexity table 2.	

Data and Safety Monitoring Table	
1. What Data and Safety Monitoring type and table is required, based on the Data Monitoring Requirement Assessment, above?	☐ Complexity Category A, table 1 ☐ Complexity Category B, table 1 ☒ Medium Complexity, table 2
2. If you believe that the required Data Safety Monitoring Table is inappropriate and will deviate from this expectation, explain here:	No

Data Safety Monitoring Parameters
1. Data Safety Monitors
Because this study involves minimal risk, non-invasive procedures, a formal Data Safety Monitoring Board (DSMB) is not required. Oversight of participant safety and data protection will be conducted by the Principal Investigator and designated study staff. The research team will be responsible for monitoring study conduct, protocol adherence, and any participant concerns.
2. Subject Safety Monitoring
Participant safety will be monitored throughout the single study interaction. Study staff will: • Observe for any signs of participant discomfort or distress during the consent process, questionnaire, or saliva collection • Remind participants that they may skip questions or stop participation at any time • Document and address any complaints, concerns, or adverse events Because procedures are non-invasive and do not involve drugs or medical devices, medical monitoring is not required.
3. Subject Stopping Rules
An individual participant's involvement will be stopped if: • The participant requests to withdraw • The participant appears unable to provide informed consent • The participant experiences distress and does not wish to continue • The participant is found to be ineligible after enrollment In these cases, no further data or samples will be collected. Participants may request destruction of their identifiable samples/data, where feasible.
4. Study Stopping Rules
The study as a whole may be paused or stopped if: • An unexpected breach of confidentiality or data security occurs that poses risk to participants • Unanticipated problems involving risks to participants or others are identified • Directed by the IRB, institutional officials, or regulatory authorities Because risks are minimal, no medical safety thresholds are required.
5. Frequency of DSMB meetings
N/A

Events and Reporting
1. Adverse Events

The following definitions and classifications have been adopted from the Winship Protocol Template:

Definition of Adverse Events (AE): Adverse event means any untoward medical occurrence associated with the use of an intervention in humans, whether or not considered intervention-related (21 CFR 312.32 (a)).

Definition of Serious Adverse Events (SAE): An adverse event or suspected adverse reaction is considered "serious" if, in the view of either the investigator or sponsor, it results in any of the following outcomes:

- Death
- Life-threatening adverse event
- Inpatient hospitalization or prolongation of existing hospitalization
- A persistent or significant incapacity or substantial disruption of the ability to conduct normal life functions, or a congenital anomaly/birth defect.
- Important medical events that may not result in death, be life-threatening, or require hospitalization may be considered serious when, based upon appropriate medical judgment, they may jeopardize the participant and may require medical or surgical intervention to prevent one of the outcomes listed in this definition. (Examples of such medical events include allergic bronchospasm requiring intensive treatment in an emergency room or at home, blood dyscrasias or convulsions that do not result in inpatient hospitalization, or the development of drug dependency or drug abuse).

Classification of an Adverse Event:
Severity of Event: For adverse events (AEs) not included in the protocol defined grading system, the following guidelines will be used to describe severity.

- **Mild** – Events require minimal or no treatment and do not interfere with the participant's daily activities.
- **Moderate** – Events result in a low level of inconvenience or concern with the therapeutic measures. Moderate events may cause some interference with functioning.
- **Severe** – Events interrupt a participant's usual daily activity and may require systemic drug therapy or other treatment. Severe events are usually potentially life-threatening or incapacitating. Of note, the term "severe" does not necessarily equate to "serious."

Relationship to Study Intervention: All adverse events (AEs) must have their relationship to study intervention assessed by the clinician who examines and evaluates the participant based on temporal relationship and their clinical judgment. The degree of certainty about causality will be graded using the categories below. In a clinical trial, the study product must always be suspect.

- **Definitely Related** – There is clear evidence to suggest a causal relationship, and other possible contributing factors can be ruled out. The clinical event, including an abnormal laboratory test result, occurs in a plausible time relationship to study intervention administration and cannot be explained by concurrent disease or other drugs or chemicals. The response to withdrawal of the study intervention (dechallenge) should be clinically plausible. The event must be pharmacologically or phenomenologically definitive, with use of a satisfactory rechallenge procedure if necessary.
- **Probably Related** – There is evidence to suggest a causal relationship, and the influence of other factors is unlikely. The clinical event, including an abnormal laboratory test result, occurs within a reasonable time after administration of the study intervention, is unlikely to be attributed to concurrent disease or other drugs or chemicals, and follows a clinically reasonable response on withdrawal (dechallenge). Rechallenge information is not required to fulfill this definition.
- **Potentially Related** – There is some evidence to suggest a causal relationship (e.g., the event occurred within a reasonable time after administration of the trial medication). However, other factors may have contributed to the event (e.g., the participant's clinical condition, other concomitant events). Although an AE may rate only as "possibly related" soon after discovery, it can be flagged as requiring more information and later be upgraded to "probably related" or "definitely related," as appropriate.
- **Unlikely to be related** – A clinical event, including an abnormal laboratory test result, whose temporal relationship to study intervention administration makes a causal relationship improbable (e.g., the event

did not occur within a reasonable time after administration of the study intervention) and in which other drugs or chemicals or underlying disease provides plausible explanations (e.g., the participant's clinical condition, other concomitant treatments). • **Not Related** – The AE is completely independent of study intervention administration, and/or evidence exists that the event is definitely related to another etiology. There must be an alternative, definitive etiology documented by the clinician.
2. External Reporting Plans
N/a

19. References

References
Haber, M., Gauguier, D., Youhanna, S., Patterson, N., Moorjani, P., Botigué, L. R., Platt, D. E., Matisoo-Smith, E., Soria-Hernanz, D. F., Wells, R. S., Bertranpetit, J., Tyler-Smith, C., Comas, D., & Zalloua, P. A. (2013). Genome-Wide Diversity in the Levant Reveals Recent Structuring by Culture. PLoS Genetics, 9(2), e1003316. https://doi.org/10.1371/journal.pgen.1003316